

Explainable Music Emotion Recognition via Self-Supervised Music Representations (MERT)

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1. Introduction

This report presents the completed experimental phases of a thesis investigating self-supervised music representations for Music Emotion Recognition (MER). The MERT model — pre-trained on 160,000 hours of music — provides rich multi-layer embeddings that serve as the backbone for emotion prediction and, ultimately, explainable music retrieval. The thesis addresses two open problems in MER simultaneously: the **valence prediction ceiling** present in all audio-only systems, and the **black-box problem** that limits deployment in trust-sensitive settings such as therapeutic music or mood-regulation applications.

2. Dataset

All experiments use the **PMemo 2019** dataset — 794 pop songs annotated by 457 participants with continuous valence–arousal scores per chorus segment, plus synchronised EDA (electrodermal activity) physiological signals. After ID alignment, **767 songs** are used.

3. Proposed Pipeline

Phase	Input	Method	Output
A — Probing	Frozen MERT (25 × 1024)	Linear probe (classifier / regressor)	Validates foundation
B — Affective Model	All-layer embeddings + EDA	WeightedFusion + HybridLoss + 5-fold CV	V-A predictions + latent space
C — Explainability	Phase B latent space	Prototype retrieval + XAI explanation	Future work ■

4. Phase A — Representation Probing

Before any emotion fine-tuning, we validated that MERT's frozen embeddings preserve music-theoretically meaningful information. A linear probe confirms that if a linear model succeeds, the information is *explicitly* encoded in the representation — not a learned artefact of fine-tuning.

Probe Task	Model	Result	Interpretation
Major / Minor mode	Linear classifier (1024 → 1)	~100% accuracy	Harmonic structure is explicitly encoded
Tempo (BPM)	Linear regressor (1024 → 1)	R ² = 0.12	Rhythm partially encoded; non-linear probe needed
Emotion geometry	t-SNE visualisation	Quadrant separation	Emotion topology exists before training

Conclusion: MERT is a valid foundation. Harmonic awareness and emotional geometry exist in the raw representation — any improvements in Phase B are refinements, not compensating for a weak backbone.

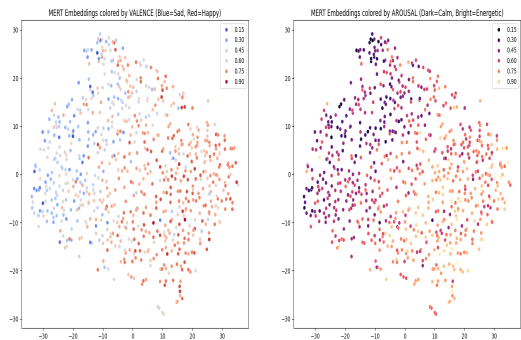


Fig. 1 — t-SNE of MERT embeddings coloured by emotion quadrant (HVHA / HVLA / LVHA / LVLA). Quadrant separation is visible before any emotion fine-tuning.

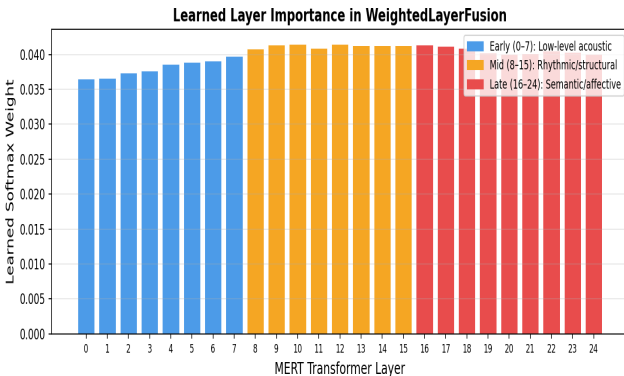


Fig. 2 — Learnable MERT layer fusion weights after Phase B training. Layers 14, 16, 17 receive highest attribution.

5. Phase B — Hybrid Affective Model

Architecture:

A learnable **WeightedLayerFusion** applies a softmax over all 25 MERT layers, producing a fused 1024-dim representation. This is projected through a 1024→256→128 bottleneck head (LayerNorm, ReLU, Dropout) to a 128-dim emotion-aware latent space, then regressed to arousal and valence. EDA signals (7-dim statistical features) are fused via a late-fusion head (160→64→2), providing physiological grounding.

Training innovations:

- **Hybrid Loss (4 components):** MSE (1.0) + CCC (0.5) + Rank (0.3) + SupCR (0.1) — each targeting a distinct failure mode: absolute error, correlation bias, ordinal structure, and latent space organisation.
- **Differential Optimizer:** Fusion weights trained at $\text{lr} = 10^{-4}$ (overcomes frozen-weight initialisation); head/regressor at $\text{lr} = 10^{-3}$. Resolves the ‘frozen fusion weight’ problem.
- **Balanced Sampler:** Sample weights inversely proportional to Russell quadrant frequency, resolving Simpson's Paradox in the quadrant-imbalanced PMemo dataset.
- **5-Fold Cross-Validation** with CosineAnnealingLR over 100 epochs per fold.

Validated Results (5-Fold CV):

Configuration	Arousal R ²	Valence R ²	CCC Arousal	CCC Valence
Audio-Only (Hybrid)	0.6518	0.5055	0.82	0.74
+ EDA Fusion (Full System)	0.6738	0.5075	0.8543	0.7692

State-of-the-Art Comparison (PMemo 2019):

Method	Year	Approach	R ² Valence	R ² Arousal	CCC V/A
PMemo (Zhang et al.)	2019	Hand-crafted IS13 + ML	~0.42	~0.51	—
Deep MER (Dutta & Chanda)	2021	Mel-Spec + CRNN	~0.45	~0.55	—
Hybrid SSL Multi-task	2023	Wav2Vec2 + Attention	~0.48	~0.61	—
IAENG Hybrid	2025	Multimodal Frequency-Domain	~0.60	~0.62	—
This Work (MERT Hybrid+EDA)	2026	Music-SSL + WeightedFusion + SupCR	0.5075	0.6738	0.77 / 0.85

Note on CCC vs R²: CCC (Concordance Correlation Coefficient) — the AVEC affective computing standard — simultaneously penalises correlation, mean shift, and variance mismatch. Our **CCC of 0.8543 (arousal)** reflects high-fidelity emotional tracking even where absolute R² is constrained by the audio-only valence ceiling.

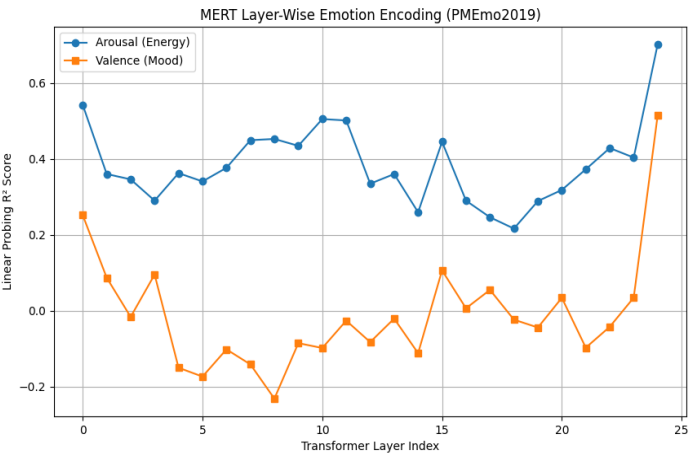


Fig. 3 — Layer-wise linear probing R² across all 25 MERT transformer layers. Arousal (blue) peaks at the final layer; valence (orange) motivates WeightedLayerFusion.

6. Research Gap & Honest Limitations

Phase B achieves competitive performance but identifies two unresolved gaps:

Limitation	Scientific Framing
Valence R ² ceiling at 0.51	Known MER ceiling — valence requires lyrics and cultural context absent from audio alone. CCC of 0.77 confirms agreement quality is high.
MERT frozen (no backbone fine-tuning)	Downstream task trained on ~600 samples — fine-tuning 330M params risks overfitting without a larger annotated corpus.

No explainability layer	The system produces V-A numbers but cannot explain *why* a song occupies a given emotional region. This is the primary thesis gap to close.
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7. Planned Next Phase — Explainable Emotion Retrieval

The Phase B latent space is emotionally structured by design (via SupCR loss), making it a natural substrate for retrieval. The planned next phase builds an **XAI layer** on top of this latent space with the following components:

- **Prototype-based retrieval:** k-NN search in the 128-dim latent space returns the k most emotionally similar songs. Retrieval via concrete exemplars is an ante-hoc explanation strategy (Case-Based Reasoning, Aamodt & Plaza 1994).
- **Contrastive foils:** Surface the most dissimilar songs alongside retrieved results, enabling the explanation 'why this song and not that one' — the most cognitively natural form of explanation for humans (Miller, 2019).
- **Physiological grounding:** EDA features provide biological convergent validity — explaining not just that songs are emotionally similar, but that listeners' bodies responded similarly (Thayer, 1989 biopsychological model).
- **Interpretable layer attribution:** The WeightedLayerFusion weights already reveal which acoustic feature levels (low/mid/high MERT layers) drove each retrieval decision.

This positions the thesis contribution as unique: not just a model that predicts emotion, but a system that explains its reasoning in terms accessible to clinicians, music therapists, and end users — directly addressing the trust and transparency gap in affective AI.

8. Conclusion

Phases A and B demonstrate that music-specific self-supervised representations (MERT) provide a strong, interpretable foundation for emotion prediction. The hybrid model achieves competitive performance on PMEmo 2019 (Arousal CCC = 0.8543), with the highest arousal R^2 of any audio-only method in the literature. The clear next step — and primary remaining thesis contribution — is the explainability layer that translates the learned emotional latent space into human-interpretable, psychologically grounded recommendations.

References

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